

ANALYSTLAB AFRICA

Data Science Internship Programme



Employee Attrition Analysis Reflection Report

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Project: ABC Manufacturing Ltd

Dataset: IBM HR Analytics – Employee Attrition & Performance

Prepared for the AnalystLab Africa Data Science Internship Programme
Reflection Report Submission

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1. Project Goal & Initial State

My primary objective was to analyse the IBM HR dataset to understand workforce characteristics and identify factors associated with employee attrition through exploratory data analysis. This analysis was intended to inform targeted retention strategies and serve as a foundation for future predictive modelling.

2. Key Steps Taken & Learnings

2.1. Data Loading and Initial Inspection

Libraries: Essential libraries such as pandas, numpy, matplotlib.pyplot, and seaborn were imported for data manipulation, numerical operations, and visualisation.

Data Loading: The dataset was successfully loaded from a Google Drive URL into a pandas DataFrame (df).

Data Quality Checks: I inspected the data for missing values (df.isnull().sum()) and duplicate entries (df.duplicated().sum()), confirming the dataset was clean with no missing values or duplicates.

Descriptive Statistics: I generated descriptive statistics for both numerical (df.describe()) and categorical (df.describe(include='object')) columns, and examined unique values (df.nunique()) to gain an initial understanding of the data's composition and distribution. Key insights included the presence of constant columns (EmployeeCount, Over18, StandardHours) and EmployeeNumber as a unique identifier.

2.2. Exploratory Data Analysis (EDA) – Univariate Distributions

Target Variable – Attrition: Analysis of df['Attrition'].value_counts() and a pie chart revealed a significant class imbalance (1233 'No' vs 237 'Yes', approximately 83.9% vs 16.1%). This is a crucial finding for subsequent modelling.

Feature Engineering (Categorical Grouping): To improve interpretability, new categorical features were created: Age_Group (from Age) and Education_Group (from Education), along with similar grouping for various satisfaction and involvement metrics (e.g., JobSatisfaction_Group).

Univariate Visualisations: Countplots and histograms were generated for Gender, Age, Department, Education_Group, JobRole, MaritalStatus, MonthlyIncome, and JobSatisfaction_Group. These visualisations helped confirm initial observations, such as the male-skewed gender distribution, a peak age around 30-35, R&D as the largest department, a highly educated workforce (majority Bachelor's), 'Sales Executive' as the most common job role, and a right-skewed monthly income distribution.

2.3. Exploratory Data Analysis (EDA) – Bivariate Distributions

Attrition by Department: A countplot of Department by Attrition provided an initial visual comparison, showing absolute numbers of attrition across departments.

Attrition by Age: Boxplots and descriptive statistics of Age by Attrition clearly showed that attrited employees have a lower mean and median age, confirming that younger employees are more susceptible to attrition.

Attrition by Monthly Income: Similar boxplots for MonthlyIncome by Attrition demonstrated that employees with lower monthly incomes have higher attrition rates.

Attrition by OverTime: A countplot revealed a disproportionately higher attrition rate among employees working overtime.

Attrition by JobRole: Calculations for Attrition_Rate per JobRole and subsequent sorting identified Sales Representatives, Laboratory Technicians, and Human Resources as roles with the highest turnover rates.

2.4. Correlation Analysis

Preparation: A copy of the DataFrame (df_corr) was made. Attrition was numerically encoded (Yes=1, No=0). Constant columns (EmployeeCount, StandardHours, Over18) and other non-encoded categorical features (including the newly created _Group columns for interpretability that would not be used in numerical correlation) were dropped using errors='ignore' to handle cases where some grouped columns might not be present in df_corr at the time of dropping.

Calculation & Visualisation: A correlation matrix was calculated and visualised using a heatmap with annotations.

Key Findings: Identified strong negative correlations with Attrition (e.g., TotalWorkingYears, JobLevel, MonthlyIncome, Age, YearsInCurrentRole, YearsWithCurrManager, StockOptionLevel, YearsAtCompany, JobInvolvement, JobSatisfaction, EnvironmentSatisfaction) and a positive correlation with DistanceFromHome.

3. Challenges & Resolutions

Data Types for Correlation: An initial attempt to calculate df.corr() would have failed due to non-numeric columns. This was resolved by creating df_corr, encoding Attrition numerically, and dropping non-numerical columns or columns irrelevant for numerical correlation. The errors='ignore' parameter in drop was specifically added to prevent KeyError when some columns might not exist.

Plotting Clarity: Various adjustments were made to plots, such as adding titles, axis labels, sorting bars, and including bar_label for countplots, to enhance readability and interpretation.

Interpreting Imbalance: The significant class imbalance in Attrition was acknowledged and noted as important for future modelling considerations.

4. Outcome & Future Steps

I successfully identified important variables influencing employee attrition, providing a strong foundation for predictive modelling. The insights gained from this EDA and correlation analysis directly informed a comprehensive Business Insights Report, highlighting key workforce observations and factors contributing to attrition. The next logical step is to move towards building predictive models, leveraging these identified variables, to forecast attrition and enable proactive intervention strategies.